

DBMS Interview Notes

RAG-friendly interview preparation material covering core concepts and explanations.

1. Database and DBMS

A database is an organized collection of related data. A Database Management System (DBMS) is software used to create, store, retrieve, update, and manage data efficiently. Examples include MySQL, PostgreSQL, Oracle, and SQL Server.

2. Advantages of DBMS

A DBMS reduces unnecessary data duplication, provides controlled access, supports data consistency, allows multiple users to work with data, and provides backup and recovery facilities.

3. Types of Databases

Common database types include relational, hierarchical, network, object-oriented, and NoSQL databases. Relational databases store data in tables and use relationships between tables.

4. Relational Database

A relational database organizes information into tables made of rows and columns. Tables can be connected using keys. SQL is commonly used to work with relational databases.

5. Keys

A primary key uniquely identifies each row in a table. A foreign key refers to a key in another table. A candidate key can uniquely identify rows and can be chosen as the primary key. A super key is any set of attributes that uniquely identifies a row. A composite key contains more than one attribute.

6. Constraints

Constraints are rules applied to table columns. Common constraints include PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, and DEFAULT. They help maintain data integrity.

7. Normalization

Normalization organizes relational data to reduce redundancy and avoid update anomalies. First Normal Form (1NF) requires atomic values. Second Normal Form (2NF) requires 1NF and removal of partial dependency. Third Normal Form (3NF) requires 2NF and removal of transitive dependency. BCNF is a stronger form of 3NF.

8. SQL

SQL, or Structured Query Language, is used to define, manipulate, control, and query relational data. CREATE, ALTER, DROP, and TRUNCATE are examples of DDL commands. INSERT, UPDATE, DELETE, and SELECT are commonly used data manipulation/query commands.

9. Joins

Joins combine related rows from multiple tables. INNER JOIN returns matching rows. LEFT JOIN returns all rows from the left table and matching rows from the right table. RIGHT JOIN does the opposite. FULL JOIN returns rows from both sides when supported.

10. Transactions and ACID

A transaction is a logical unit of database work. ACID stands for Atomicity, Consistency, Isolation, and Durability. Atomicity means all operations of a transaction succeed or the transaction is rolled back. Consistency preserves valid database rules. Isolation controls interaction between concurrent transactions. Durability keeps committed changes.

11. Indexing

An index is a data structure that can make data retrieval faster. Indexes are useful for frequently searched columns, but they require storage and can add overhead to INSERT, UPDATE, and DELETE operations.

12. Views

A view is a virtual table based on a query. It can simplify complex queries and restrict access to selected data. A materialized view stores query results physically and can be refreshed.

13. ER Model

An Entity-Relationship model represents entities, attributes, and relationships. Entities represent objects such as Student or Course. Relationships describe how entities are connected.

14. Data Dictionary

A data dictionary stores metadata about a database, such as table names, column names, data types, constraints, and other structural information.

15. Database Security

Database security protects data against unauthorized access, modification, and loss. Common practices include authentication, authorization, roles, privileges, encryption, backups, and auditing.